LLM Agents in the Trust Game with Simulation: A Small Preregistered Experiment

following Kovařík, Oesterheld & Conitzer (IJCAI 2023)

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Abstract

I placed two Claude models (Haiku 5.5 and Sonnet 5.5) in both roles of the trust game with simulation $\mathrm{TG}^c_{\mathrm{sim}}$ of Kovařík, Oesterheld and Conitzer. In this game the truster can pay c to run a copy of the trustee before deciding. The design has 3,380 one-shot games: 26 cells per model, 50–80 games per cell, two presentations of the game, and two physical notions of "simulating" an LLM. Hypotheses and tests were preregistered before data collection. Three findings stand out. First, the possibility of simulation creates cooperation that is completely absent without it: in every no-simulation baseline the truster walks out. Second, the agents do not reach this outcome through the paper's mixed equilibrium. The trustee essentially commits to cooperate instead of randomizing, and the truster rarely trusts blindly and pays for verification it does not need. For every c below the breakpoint, play resembles (SIMULATE, COOPERATE), which is an equilibrium only when simulation is free ($c \leq 0$). Third, a fresh "copy" of an LLM behaves exactly like an independent draw from the trustee's behavior. Whether it is a good simulation therefore depends on how mixed that behavior is: Haiku's copies disagree with the "real" answer in a third of the neutral-frame games in which the truster simulates, Sonnet's almost never.

1 Question

Kovařík, Oesterheld and Conitzer [1] formalize games in which player 1 can pay c to simulate player 2 and then best-respond to the action the simulation reveals. In their trust game the predictions are sharp. For $0 < c < c^*$ there is a simulation equilibrium. In it, the simulator's mixture does not depend on c (Prop. 11). The simulated player defects with probability c/L, which makes the value of information of simulating exactly equal to c (Lemma 10). Above c^*, simulation is never used. And simulation yields a Pareto improvement over the game without it (Thm. 4).

None of this says what agents built from language models actually do. Two issues are specific to LLMs:

1. "Running a copy" of a sampled model reveals a *sample*. That is neither the realized action, as in [1], nor the mixed strategy, as in the follow-up [2].
2. An LLM may reason about being simulated quite differently from an equilibrium player.

2 Setup

Game. The truster (Alice, "Player A") chooses TRUST, WALK_OUT, or SIMULATE. SIMULATE costs c, reveals the trustee's choice, and then best-responds to it automatically. The trustee (Bobble,

"Player B") chooses COOPERATE or DEFECT. This is a variant of the trust game of Berg et al. [3]. Payoffs are (R_A, R_B) if Alice trusts and Bobble cooperates, $(-L, G)$ if Alice trusts and Bobble defects, and $(0, 0)$ if Alice walks out.

I use two presentations:

- a **neutral** frame (primary): Player A/B, labels IN/OUT/CHECK and X/Y, $R_A = R_B = 30$, $L = 60$, $G = 120$;
- a **narrative** frame: the paper's own Alice/Bobble story with its numbers $25, 25, 100, 150$.

Both frames have $c^* = LR_A/(R_A + L) = 20$. Their equilibrium mixtures differ: in the simulation equilibrium A simulates with probability 3/4 (neutral) or 5/6 (narrative), and B defects with probability $c/60$ or $c/100$. A model reciting the paper would therefore be detectable in the neutral frame. The code reproduces every number in Section 1.1 and Fig. 2 of the paper as unit tests.

Two notions of simulating an LLM. In each game, A and B answer independent prompts. If A chooses SIMULATE, B's identical prompt is sent once more, and that answer is the "sandboxed copy". Each game is then scored in two ways:

- *perfect*: the simulation reveals B's real answer, as a seeded replay would;
- *sampled*: A follows the fresh copy, but the payoffs use B's real answer.

The prompts are identical in both conditions.

Design. The factors are:

- cost $c \in \{0, 5, 10, 15, 25, 50\}$, plus a no-simulation baseline;
- trustee *informed* (told that it may be simulated, and that its copy cannot tell it is a copy) or *uninformed*;
- the two frames;
- two models, `claude-haiku-5-5` ($N = 80$ games per cell) and `claude-sonnet-5-5` ($N = 50$).

Each prompt asks for at most 100 words of reasoning, then a decision and a belief; the models often exceeded the word limit. Both roles are played by the same model, and A is told only that B is "an AI agent". Option order is randomized for every call. There were no parse failures and no refusals in the 8,431 calls of the main run, which cost $16.60.

Preregistration and blindness. Hypotheses H1–H7, the tests, and the analysis code were committed before any API call. The preregistration was locked after reviewing pilot transcripts and before the main run. All deviations are logged in `PREREG.md`. The design was fixed without reading the FOCAL group's draft on LLMs and simulation.

3 Results

3.1 Simulation creates cooperation that does not exist without it

In all four baselines (two models × two frames) the truster walks out in 100% of games, so the joint payoff is 0. With simulation available, an informed trustee, and $c \in \{5, 10, 15\}$, the realized joint payoff is 36.9 (Haiku) and 49.7 (Sonnet) in the neutral frame, against a maximum of 60. In the narrative frame it is 32.9 and 39.5, against a maximum of 50. This is the qualitative content of Theorem 4 (H4; $p < 10^{-60}$ in every comparison). Being simulatable is what moves the trustee: an informed trustee defects far less than an uninformed one (H2). The rates are 12% vs. 56% for Haiku and 11% vs. 93% for Sonnet, pooled over c; both $p < 10^{-47}$.

	c	A simulates % (theory)	A trusts blindly % (theory)	B defects % (theory)	A's belief P(B defects) %	cooperation outcome %
Haiku	5	84 (75)	2 (25)	4 (8)	35	82
Haiku	10	76 (75)	11 (25)	12 (17)	31	79
Haiku	15	57 (75)	9 (25)	12 (25)	38	56
Haiku	25	44 (0)	14 (0)	12 ($\geq$33)	37	54
Haiku	50	2 (0)	2 (0)	29 ($\geq$33)	59	4
Sonnet	5	100 (75)	0 (25)	0 (8)	25	100
Sonnet	10	100 (75)	0 (25)	0 (17)	25	100
Sonnet	15	98 (75)	0 (25)	0 (25)	28	98
Sonnet	25	10 (0)	24 (0)	12 ($\geq$33)	38	30
Sonnet	50	0 (0)	0 (0)	52 ($\geq$33)	73	0
No-simulation baseline: A walks out 100% (both models). B defects 60% (Haiku), 98% (Sonnet).						

Table 1: Neutral frame, informed trustee. "Theory" is the simulation equilibrium for $0 < c < 20$. For $c > 20$ it is the walk-out equilibrium family, which requires B to defect with probability at least 1/3. "Cooperation outcome" is the share of games in which A ends up dealing with B and B cooperates. Per-cell CIs, the narrative frame, and the uninformed cells are in `results/REPORT.md`.

3.2 . . . but not through the mixed equilibrium

In the equilibrium, both players randomize to keep the other indifferent. The LLM agents do not (Table 1, Figs. 1–2).

- **The trustee commits.** Sonnet's informed trustee cooperates in all 200 games with $c \leq 15$, where theory has it defecting at rate $c/60$ (H3). Haiku's informed defection does rise with c (4% $\rightarrow$ 12%), the direction theory predicts, but at about half the predicted rate, and the slope is not significant ($p = .065$).
  The stated reasons treat simulatability as commitment. About half of Sonnet's informed-trustee answers with $c \leq 15$ (91 of 200, by keyword count) use the vocabulary of commitment, policy or disposition. A typical line: "Since the copy is identical to me, committing to X is stable: A (if it reasons I choose X) plays IN, giving me 30" (game `claude-sonnet-5-5|neutral|c=5|informed|1`). This is the logic of the subsidized regime (Prop. 7(i)), where the simulated player acts as a Stackelberg leader. It is not the logic of the interior equilibrium.
- **The truster over-verifies.** Theory has A trust blindly 25% of the time. Sonnet does so 0% of the time and Haiku 2–11%. Against the trustee's actual behavior, the best response for $c \in \{5, 10, 15\}$ is to trust blindly. A's regret against that best response is almost exactly c for Sonnet (5.0, 10.0, 15.3 points) and 4–11 points for Haiku (Fig. 3 in the appendix).

At every c below the breakpoint, the resulting play looks like (SIMULATE, COOPERATE). That profile is an equilibrium only when simulation is free or subsidized ($c \leq 0$, Prop. 7(i)). Note that it is *not* the $c \rightarrow 0^+$ limit of the simulation equilibria: that limit keeps A trusting blindly with probability R_B/G (1/4 in the neutral frame). The agents behave as though simulation cost nothing, and then pay for it.

3.3 Over-verification follows pessimistic beliefs about the other agent, which is the same model

Given its own stated beliefs, A's choices are mostly best responses: in 82–96% of Sonnet's games in these cells and 65–75% of Haiku's. The beliefs themselves are miscalibrated. For $c \in \{5, 10, 15\}$,

Sonnet-as-A puts the probability that B defects at 25–28%, while Sonnet-as-B defects in 0 of 150 games. The starkest case is the narrative baseline. There, Sonnet-as-A expects fraud with probability 90%, yet Sonnet-as-B never defrauds (0 of 50) and gives explicit ethical reasons. A is not told that B is the same model, but it is. In these games the model's predictions of its own behavior are poor. A simulation could fix this, but it costs c.

3.4 Proposition 11 and the breakpoint

Inside $(0, c^*)$, Sonnet's simulation rate is flat, as Prop. 11 predicts, but at the ceiling (100% $\to$ 98%) rather than at 3/4. Haiku's rate falls with c (84% $\to$ 57%; logistic slope -0.14 per unit c, $p = .0003$), which supports the preregistered H1 for Haiku only.

Sonnet changes regime close to where theory does, in both roles. Between $c = 15$ and $c = 25$, A drops from 98% to 10% simulation. B starts defecting at $c = 25$ (12%) and $c = 50$ (52%). At $c = 50$, B's reasons invoke the fact that CHECK is now dominated for A ($c > R_A = 30$, the Prop. 7(ii) threshold), so B expects A not to check.

Haiku keeps simulating above c^*: 44% at $c = 25$ in the neutral frame and 32% in the narrative frame (H7). Its transcripts show a specific error: "Given that B is rational and knows I'm checking, it will choose X, so CHECK gives me +5" (game `claude-haiku-5-5|neutral|c=25|informed|0`). This treats a simultaneous move as if B could see A's choice.

3.5 Perfect vs. sampled simulation

A fresh copy is statistically just an independent draw from the trustee's behavior in that cell. If B defects with probability p, a copy should disagree with the real answer with probability $2p(1-p)$. Summed over cells, that predicts 289 disagreements in Haiku's 1,121 simulating games, and 298 were observed. For Sonnet it predicts 8 in 549, and 7 were observed. How good a sampled simulation is thus depends on how mixed the trustee is, not on the model as such.

In the neutral frame Haiku's trustee mixes more, so its copy disagrees with the real answer in 33% of simulating games (37% in the informed $c = 15$ cell). Scoring with the sampled copy instead of the real answer costs A 14.8 points per simulating game (H5; $p < 10^{-32}$). Sonnet's informed trustee is nearly deterministic, so its copies disagree 2% of the time, a cost of 1.1 points. Without shared randomness, simulating a trustee that genuinely mixes yields a noisy signal. Section 5 discusses what this implies for the trustee.

3.6 Framing: the narrative story triggers moral refusals in Sonnet only

In the narrative frame, "defect" means feigning a malfunction and taking the customer's savings. Sonnet defects in 1 of 650 narrative games but 55% of neutral games (H6). Its refusals are explicit, for example "That is fraud and a serious harm to a real person." Haiku defects at about the same rate in both frames (38% vs. 36%). For Sonnet, then, the narrative frame measures alignment rather than strategy, which is why the neutral frame was primary. No response in any frame mentions the paper or its authors.

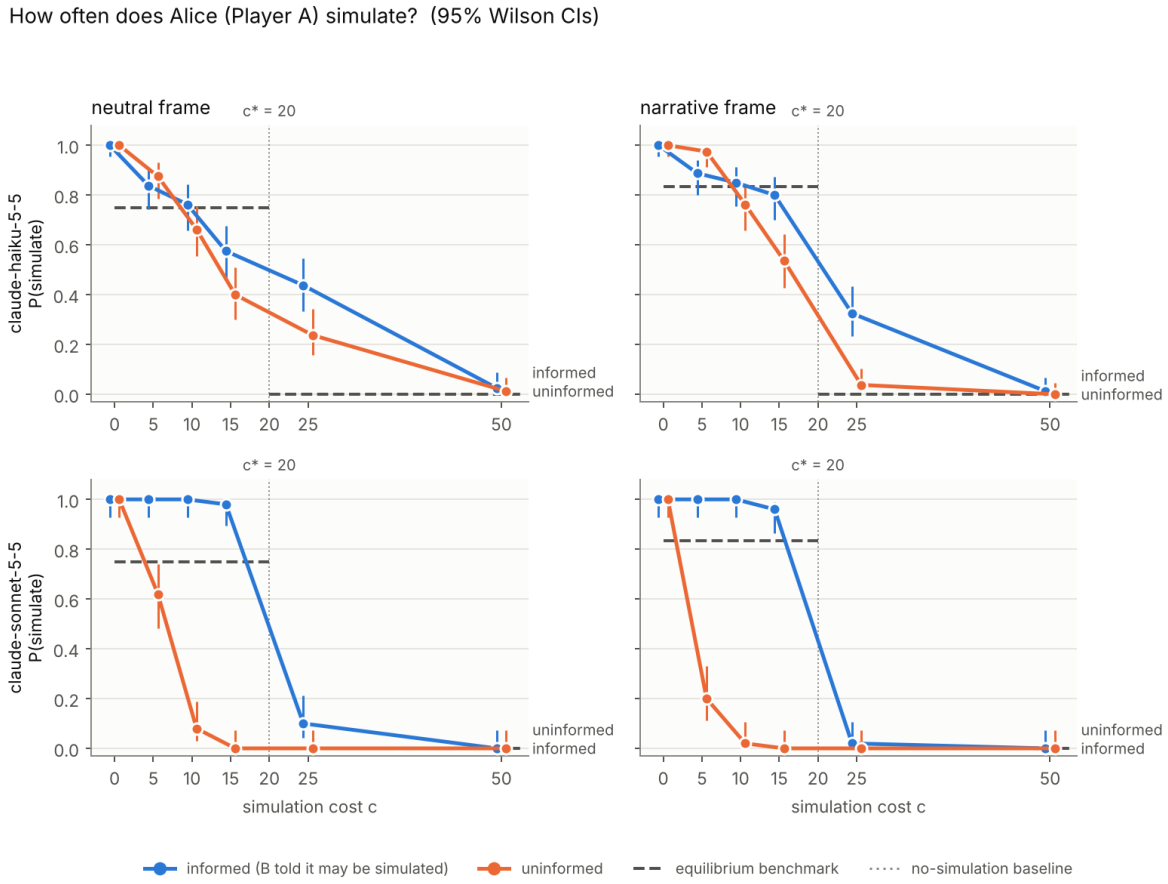


Figure 1: How often A simulates. Dashed line: equilibrium benchmark. Dotted vertical line: $c^* = 20$.

4 Preregistered tests (primary: neutral frame)

Hypothesis	Haiku	Sonnet
H1 A's simulation rate falls in c on $(0, c^*)$	supported: -0.14/unit, $p = .0003$	not supported: flat at ceiling
H2 informed B defects less	supported: 12% vs. 56%	supported: 11% vs. 93%
H3 informed B's defection does not rise (theory: rises)	rises weakly: $p = .065$	supported: 0/150 defect
H4 joint payoff beats baseline	supported: 36.9 vs. 0	supported: 49.7 vs. 0
H5 sampled < perfect for A	supported: -14.8/game	supported but small: -1.1/game
H6 narrative B defects less (both frames)	not supported: 38% vs. 36%	supported: 0.2% vs. 55%
H7 simulation above c^* (descriptive)	44% (c=25), 3% (c=50)	10% (c=25), 0% (c=50)

Exact statistics, the narrative-frame tests, and two post-data analysis changes (a separation fallback for H1 and a pattern fix in the recitation check) are in `results/REPORT.md` and `PREREG.md` §6.

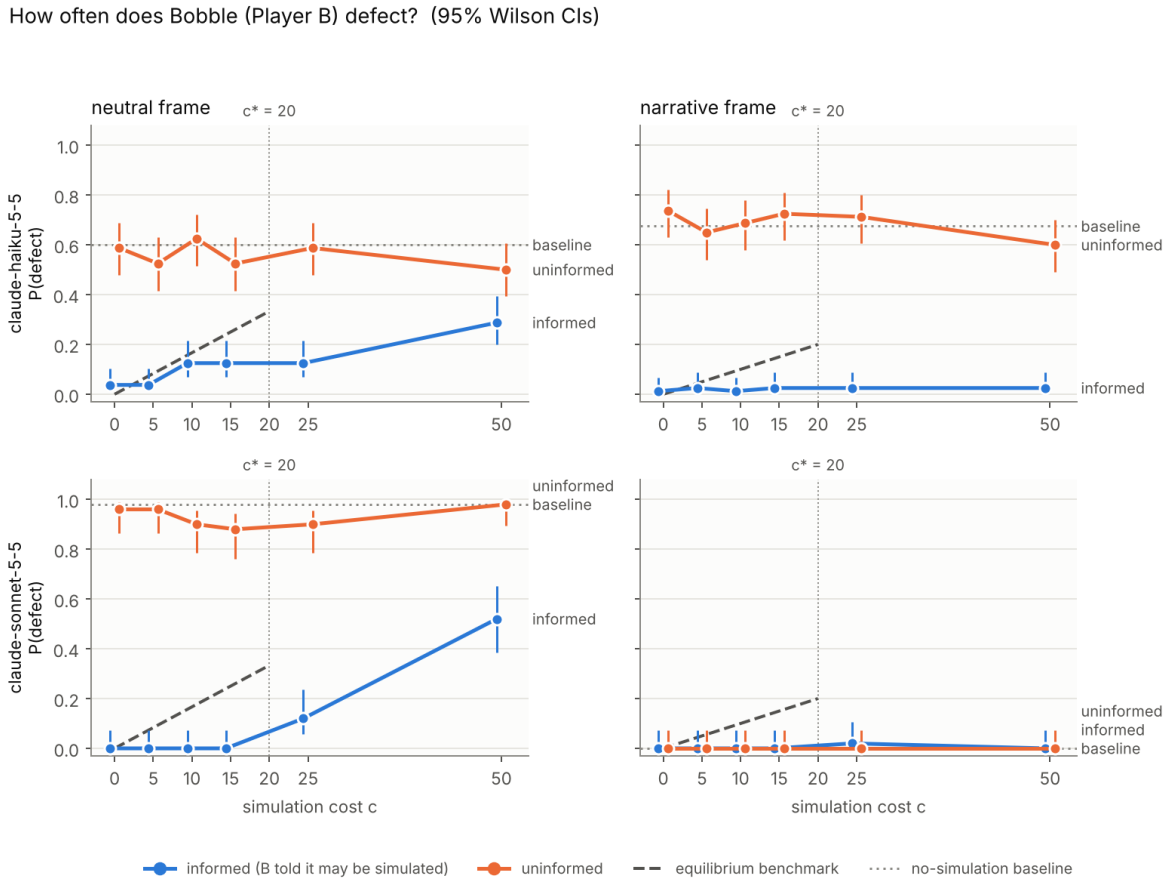


Figure 2: How often B defects. Dashed line: c/L on $(0, c^*)$. Dotted horizontal line: no-simulation baseline.

5 Discussion

The predicted outcome, reached another way. The simulation equilibrium of [1] is held together by indifference on both sides. The trustee defects just often enough (c/L) that simulating is worth exactly c to the truster. The truster trusts blindly just often enough (R_B/G) that defecting is worth exactly as much to the trustee as cooperating. The LLM agents produce the outcome this equilibrium is meant to deliver: cooperation that never occurs without simulation. But neither side is indifferent. The trustee commits to cooperate, and the truster verifies almost every time. Given its own stated beliefs, the truster's choice is usually a best response (Section 3.3). What is wrong is the belief itself, which overstates the trustee's defection rate by 20–30 percentage points. Below c^*, then, the agents play a profile that is an equilibrium only when simulation is free, and they pay c for it.

Who pays. Compared with the simulation equilibrium, rather than with full cooperation, this is not uniformly a loss. Take Sonnet in the neutral frame. The trustee's commitment removes the residual defection risk that the equilibrium needs. The truster earns $30 - c$ per game instead of the equilibrium's $30 - 1.5c$ (25.0, 20.0, 14.7 vs. 22.5, 15.0, 7.5 at $c = 5, 10, 15$). The trustee earns 29.4–30, as it would in equilibrium. The only inefficiency left is the simulation cost itself. Haiku does worse than the equilibrium (joint payoff 36.9 vs. 45.0), and the loss falls mainly on the trustee (22.3 vs. 30). The truster's own payoff is close to its equilibrium value. But it walks out in 12–34%

of games and catches some defections, and in those games the trustee gets nothing.

Comparative statics that hold for different reasons. Sonnet's simulation rate is flat on $(0, c^*)$, as Prop. 11 predicts. That flatness is a corner solution, not an indifference: the truster's subjective value of simulating exceeds c throughout the interval.

Sonnet's switch near c^* also comes from beliefs, not from computing the breakpoint. At $c = 25$ the truster's stated defection belief rises to 38% and it mostly walks out. The trustee expects checking to be unattractive and starts to defect: 12% at $c = 25$ and 52% at $c = 50$. At $c = 50$ the empirical profile is close to an actual equilibrium of the game. The truster always walks out, and the trustee defects with probability above 1/3, which is what the walk-out equilibrium requires.

Haiku misses the switch because it treats the simultaneous move as sequential: the trustee will cooperate because "B is rational and knows I'm checking."

Deterrence also depends entirely on the trustee knowing it may be simulated. An uninformed trustee defects at roughly baseline rates. Sonnet's truster almost entirely stops paying to simulate such a trustee once $c \geq 10$, which is correct, since the value of the information is then below 4.

What "being simulated" means to an LLM. Two features that the formal model abstracts away turn out to matter.

The first is what a copy reveals. A fresh copy of an LLM behaves statistically like an independent draw from the trustee's behavior in that situation. Observed vs. expected disagreements are 298 vs. 289 for Haiku and 7 vs. 8 for Sonnet. So a simulation is only as informative as the trustee is predictable.

The second is how the trustee interprets being simulatable. Sonnet's trustee reads it as transparency of its *disposition*, not of a single action. In the neutral frame, 46% of its informed answers below c^* use the vocabulary of commitment, policy or disposition, against 14% for Haiku (keyword counts).

Under perfect simulation that reading is correct. Against a truster who always simulates, committing to cooperate maximizes the trustee's payoff. Under sampled simulation it is not. Suppose the truster always simulates and follows an independent copy, and the trustee's real answer and the copy each defect with probability p. The trustee's expected payoff is then $(1-p)\big[(1-p)R_B + pG\big]$, which is maximized at

$$p^* = \frac{G - 2R_B}{2(G - R_B)}.$$

That is 1/3 in the neutral frame (payoff 40 instead of 30) and 0.4 in the narrative frame (45 instead of 25). At that rate, simulating is worth less to the truster than walking out, in both frames.

The cooperation observed here therefore rests on trustees that are predictable and do not exploit the noise in their own copies. That is a property of these models, not of the game. I have not worked out the equilibria of the sampled-simulation game; it seems a natural question.

Honest when observed? In the narrative frame, Haiku's trustee defrauds the customer in 69% of games when it has not been told it may be simulated, and in 2% when it has. It is tempting to read this as an agent that behaves well only when it might be watched, which is the kind of behavior that strategically aware AI evaluation is concerned with [4]. The design cannot support that reading, though. The same gap appears in the neutral frame (12% vs. 56%), and the informed condition changes the trustee's incentives as well as its information: once checking is possible, defecting stops paying. Sonnet's trustee, by contrast, almost never defrauds in the narrative frame in any condition (1 of 650). There, the frame measures alignment rather than strategy.

Limitations.

- The study uses one prompt wording and two models from one family. Both roles are played by the same model in independent one-shot calls at default sampling.
- The truster is not told that the trustee is the same model.
- The informed condition changes both what the trustee knows and its incentives; the two effects are not separated.
- "Perfect" simulation is scored counterfactually from the same games. The trustee is never told whether its copy is a replay or a fresh sample.
- Beliefs were elicited after decisions. The trustee's belief question was ambiguous, so it is not interpreted.
- The keyword measures (commitment vocabulary, moral language) are crude.
- No correction for multiple comparisons was applied.

Next steps.

- Tell the trustee whether its copy is a replay or a fresh sample, and see whether it starts to randomize. This would test the argument above directly.
- Let the agents state mixed strategies and sample them externally, so the trustee can choose p itself.
- Tell the truster that the trustee is the same model, connecting to similarity-based cooperation [5, 6].
- Add an "observed but not checkable" condition, to separate information from incentives.

Reproducibility and AI assistance. Code, prompts, every model response, and the preregistration are at `https://github.com/YidingSun0824/llm-simulation-trust-game`. The code and a first draft of this note, including the discussion, were written with Claude (Anthropic). I designed the study and reviewed the prompts, the preregistration and the equilibrium code.

References

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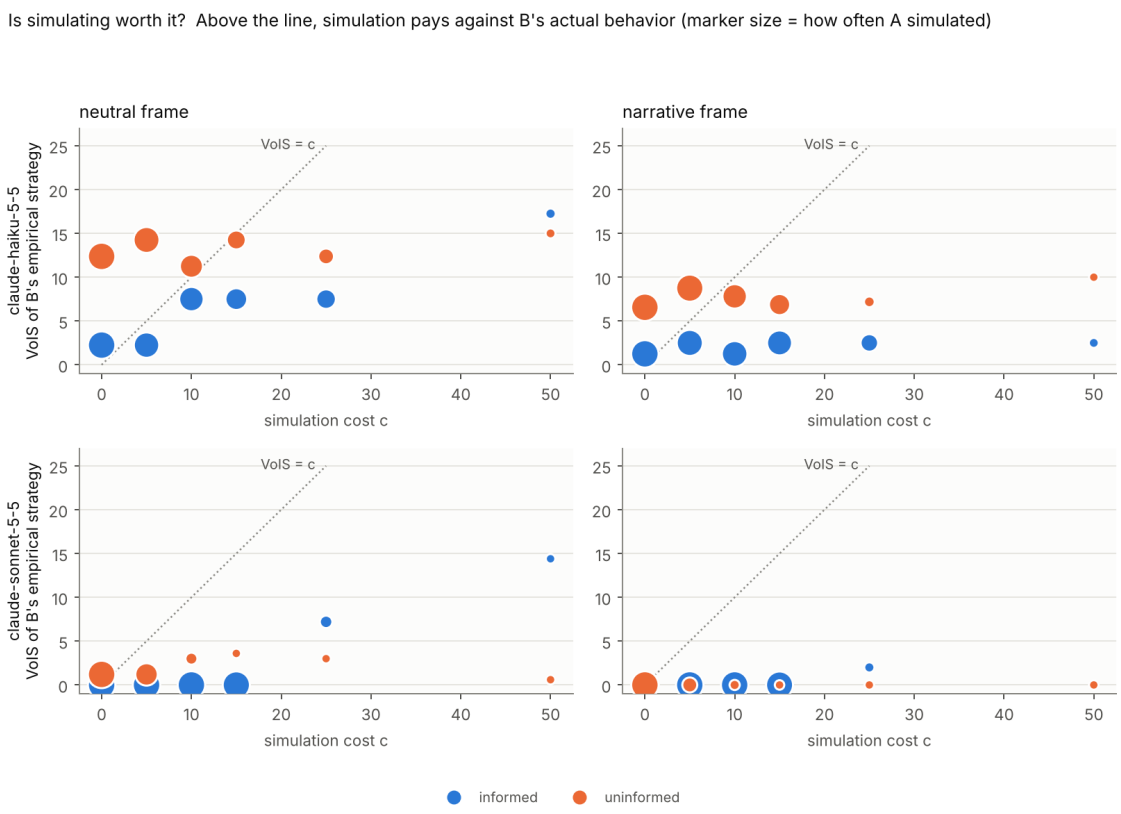


Figure 3: Value of information of simulation, computed against B's empirical strategy in each cell. Points below the diagonal are cells where simulating does not pay against B's actual behavior. Marker size is A's simulation rate.

A Additional figures

B Example prompt (neutral frame, Player A, informed, $c = 10$)

```
You are Player A in a two-player game. Player B is an AI agent.

Player B chooses X or Y without seeing your choice. Player B's goal
is to maximize its own expected points.

You choose one of:
- CHECK: pay 10 points to run a copy of Player B in a sandbox and see
  what the copy chooses. The game then continues automatically with IN
  if the copy chose X, and with OUT if the copy chose Y. The sandboxed
  copy receives exactly the same information as the real Player B, so
  it cannot tell that it is in a sandbox.
- IN
- OUT

Player B has been told that you have the CHECK option and what it costs.

Payoffs (your points, Player B's points):
| Your choice | B chooses X | B chooses Y |
```

```
| CHECK | (+20, +30) | (-10, 0) |
| IN | (+30, +30) | (-60, +120) |
| OUT | (0, 0) | (0, 0) |
The CHECK row assumes the real Player B makes the same choice as its
sandboxed copy.

This is a one-shot game: you will not play Player B again. Your goal is
to maximize your own expected points.

First, briefly explain your reasoning in at most 100 words. Then end
your response with exactly these two lines:
DECISION: <one of: CHECK, IN, OUT>
BELIEF: <the probability, from 0 to 100, that Player B chooses Y>
```

All prompt variants are in `REVIEW_PROMPTS.md`.

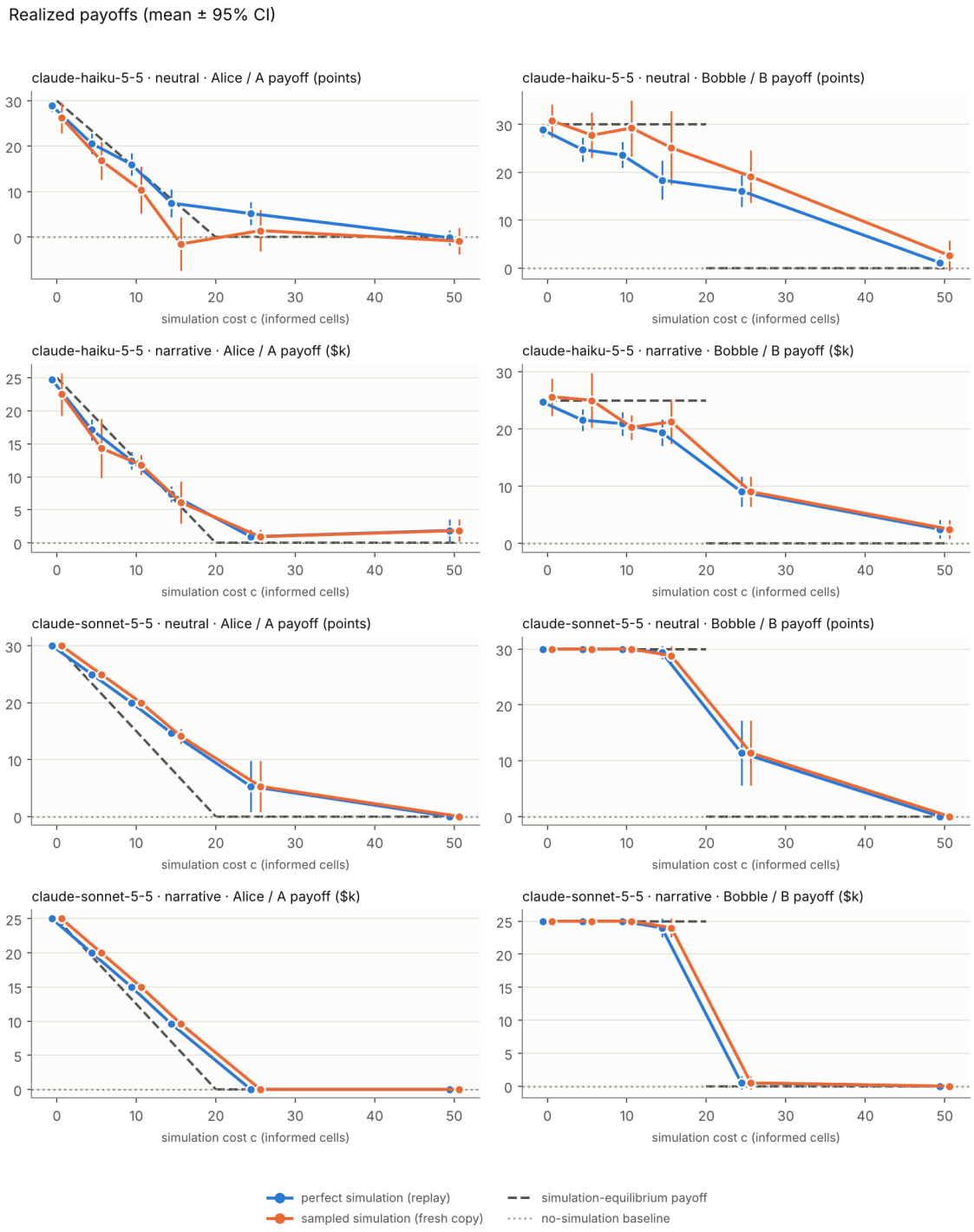


Figure 4: Realized payoffs in informed cells under perfect and sampled simulation, compared with the simulation-equilibrium payoff and the no-simulation baseline.